



M002 · Turning (Advanced)

Topic: Turn at Walls · **Course:** Maze Madness · **Level:** Advanced · **Time:** about 55 minutes

The rule: wall ahead, turn. No wall, move forward.

1 · Trace It 🧠

```
while True:
    if agent.detect(BLOCK, FORWARD):
        agent.turn(RIGHT_TURN)
    else:
        agent.move(FORWARD, 1)
```

Say what the agent does each loop.

2 · Read the API

Code	Means
<code>agent.detect(BLOCK, FORWARD)</code>	wall in front?
<code>agent.turn(RIGHT_TURN)</code>	turn right
<code>agent.turn(LEFT_TURN)</code>	turn left
<code>agent.move(FORWARD, 1)</code>	go forward 1

Write one line: turn left if a wall is on the LEFT.

3 · Spot the Bug 🐛

```
# A
if agent.detect(BLOCK, FORWARD):
    agent.turn(RIGHT_TURN)
    agent.move(FORWARD, 1)
```

What is wrong in A?


```
# B
if agent.detect(BLOCK, LEFT):
    agent.turn(RIGHT_TURN)
else:
    agent.move(FORWARD, 1)
```

B should turn at a front wall. What is wrong?

4 · Fix It

Rewrite B so it turns left when a wall is in front, else moves forward.

5 · Write Full Code

Loop forever. Wall ahead? Turn right. No wall? Move forward 1.

6 · Chat Command

`player.on_chat("go", fn)` runs `fn` when you type `go` in chat.

Write code so typing `go` starts your maze loop.

7 · Show Your Work 📷 🎥

Open your homework world. Get the agent out with your turn code.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.